

Tirupati Surface-Water & Monitoring — Project Report

Site: Tirupati district & surrounds, Andhra Pradesh, India **Sensor:** Sentinel-1 C-band SAR (RTC GRD, amplitude) **Date:** 2026-07-16 · **Status:** Operational — implemented, verified, run end-to-end

Abstract

This report presents a near-real-time surface-water, drought, and flood change-detection analysis for the Tirupati district using **Sentinel-1 Synthetic Aperture Radar (SAR)** amplitude imagery. Because radar images through cloud and darkness, the workflow keeps operating through the South-West monsoon, when optical sensors fail. A 16-date Sentinel-1 time series (2026-01-06 → 2026-06-30) is processed by a single deterministic pipeline that classifies **open water** from backscatter and detects **new inundation** against a dry-season baseline, using three complementary detectors: a fixed decibel threshold, a per-scene adaptive Otsu threshold, and a self-supervised deep change detector (**SAFNet**). Detection is made **dual-polarization** ($VV \wedge VH$) to reject dry smooth surfaces, and outputs are exported as georeferenced GeoTIFFs. Over the analysis window the pipeline resolves the district's hydrological cycle — confident open water falling from $\sim 1.4 \text{ km}^2$ in January to a $\sim 0.5 \text{ km}^2$ minimum by late June, then rebounding to $\sim 2.4 \text{ km}^2$ at monsoon onset — a directly usable drought/drawdown signal.

Area of Interest

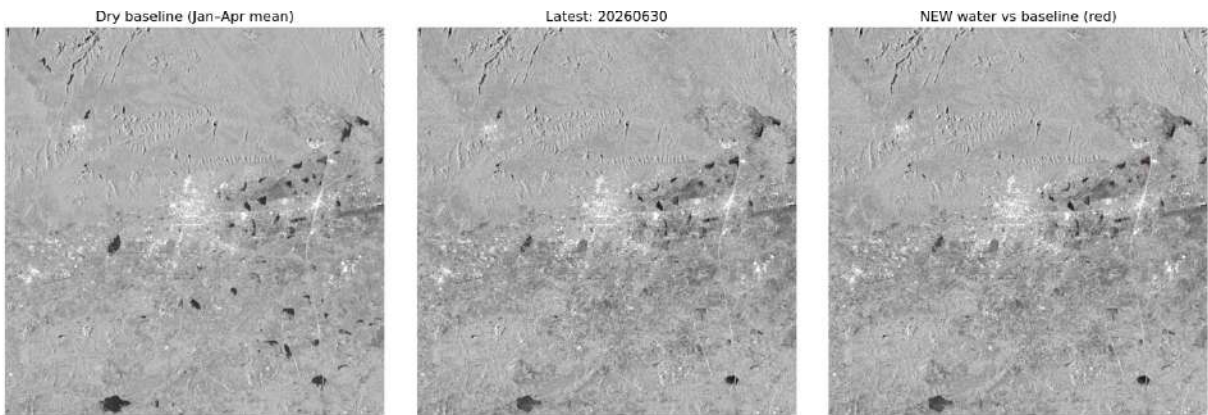
Parameter	Value
Location	Tirupati district & coastal Andhra Pradesh
Bounding box (W,S,E,N)	79.298, 13.498, 79.552, 13.752 (EPSG:4326)
Extent	$\sim 27 \times 28 \text{ km}$ tile centered on Tirupati city
Projection	EPSG:32644 (UTM zone 44N)
Terrain / hydrology	Semi-arid plateau; irrigation tanks and reservoirs subject to strong seasonal drawdown

The AOI is deliberately compact, so the full multi-detector pipeline (including the CNN) runs at district scale in minutes.

Pre-processing and Auxiliary Data

Component	Detail
Product	Sentinel-1 RTC GRD (Radiometrically Terrain-Corrected γ^0 , linear backscatter), IW mode.
Source	Microsoft Planetary Computer (sentinel-1-rtc collection), signed access.
Scenes	16 acquisitions, 2026-01-06 → 2026-06-30 , ~12-day cadence (dry winter → pre-monsoon → monsoon onset).
Polarizations	Dual-pol VV + VH (~823 MB total in data/Tirupati/).
Native grid	10 m pixels; nodata = -32768.
Grid alignment	Every scene (VV and paired VH) is warped onto one common ~40 m reference grid (WarpedVRT, downsample $\times 4$), co-registering all dates and guarding array stacking against footprint drift.
Radiometric prep	Average resampling on read (cheap speckle reduction); linear $\gamma^0 \rightarrow$ decibels ($10 \cdot \log_{10}$); nodata / non-positive samples masked to NaN so invalid pixels are never counted as water.

No DEM or external mask is required for the amplitude workflow; the reference-grid transform is retained so all raster outputs are georeferenced (EPSG:32644).



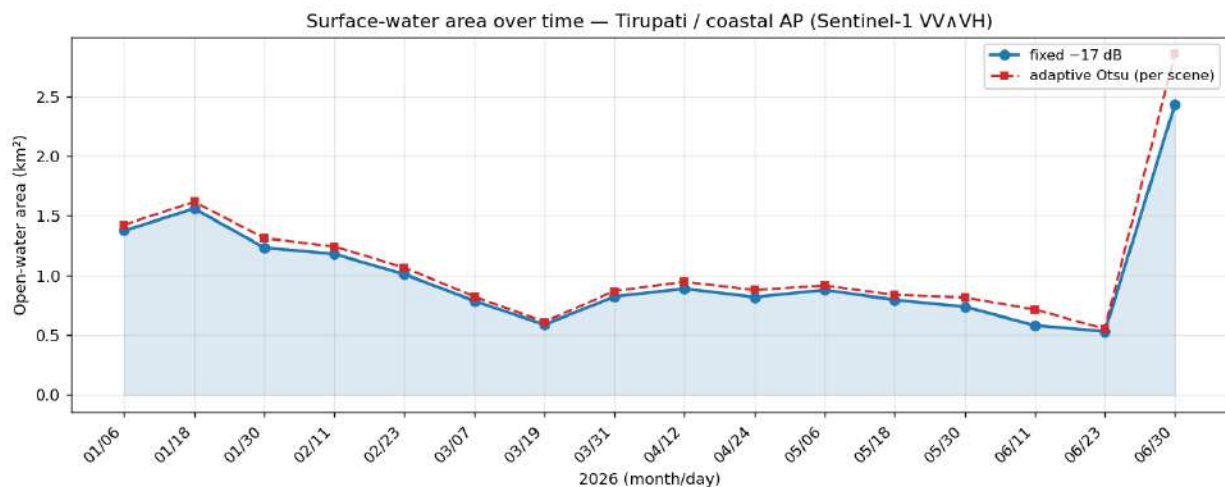
Methodology and Techniques

Dual-polarization water primitive. A pixel is classed **open water** only when dark in **both** polarizations — $VV < -17$ dB and $VH < -24$ dB. Requiring co- and cross-pol darkness rejects smooth dry surfaces (roads, runways, dry sand) that fool a VV-only test, cutting false positives.

Three complementary detectors:

1. **Fixed -17 dB VV threshold** — the classical, well-understood baseline.
2. **Per-scene adaptive Otsu** — Otsu computed on the *dark tail* of each scene's histogram (open water is ~ 0.1 % of pixels, so whole-histogram Otsu would over-detect), clamped to ± 3 dB around -17 dB so noise cannot drift the cut.
3. **SAFNet — Siamese Adaptive Fusion Network** (self-supervised, no labels): a shared-weight multi-scale encoder embeds both dates; per-scale absolute difference features are combined by a learned **per-pixel softmax attention** into a change probability. Trained on **reconstruction** (learn SAR texture) + **synthetic copy-paste change** (pasted content = change, added speckle = no change), so it learns SAR-noise invariance rather than the -17 dB rule.

Change detection. New water = *open water now (dual-pol) and dry in the baseline*. Both the threshold and SAFNet change masks are **speckle-cleaned** (morphological opening + removal of connected components < 5 px).



Core SAR Processing Chain

(This track uses SAR **amplitude** (GRD), so there is no interferometric phase — the "core chain" is amplitude classification + change detection, the amplitude analogue of an InSAR chain.)

Discover glob VV scenes, parse YYYYMMDD, sort chronologically

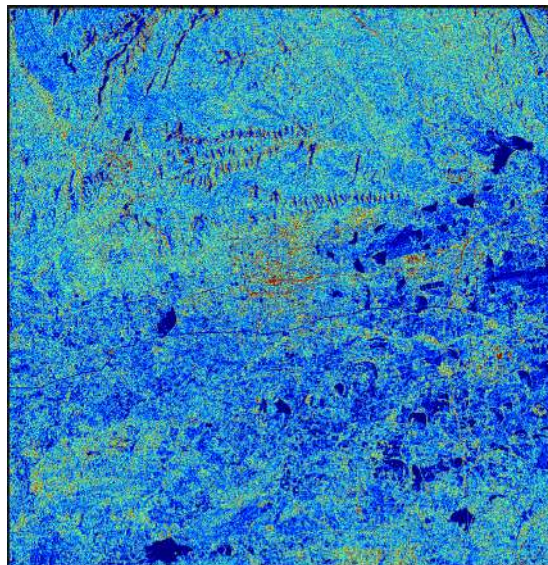
|

Align + dB WarpedVRT → common ~ 40 m grid → to_db → nodata/NaN (VV and VH)

|
 Water mask dual-pol: $VV < -17 \text{ dB} \wedge VH < -24 \text{ dB}$ (per scene)
 |
 Time series count water px $\rightarrow$ km² (fixed cut + adaptive Otsu) $\rightarrow$ figure 1
 |
 Baseline mean dB of first ~8 (dry) scenes vs latest scene
 |
 New water (A) water now (dual-pol) $\wedge$ dry in baseline $\rightarrow$ despeckle $\rightarrow$ figure 2 + GeoTIFF
 |
 SAFNet (B) self-supervised train $\rightarrow$ change probability $\rightarrow$ Otsu cut
 $\rightarrow$ changed $\wedge$ water now $\rightarrow$ despeckle $\rightarrow$ figure 3 + GeoTIFFs
 |
 Validate SAFNet scored vs threshold (PROXY) or vs REF_WATER (ground truth)

SAFNet architecture: multi-scale Siamese encoder (channels $1 \rightarrow 16 \rightarrow 32 \rightarrow 64$, max-pool between scales); a decoder used only for reconstruction pre-training; and an adaptive-fusion head that projects each scale's $|fA - fB|$, upsamples, and combines them via per-pixel softmax attention into a change logit. Losses: reconstruction (MSE) + synthetic-change (BCE), optimised jointly.

Water-area conversion uses $\text{px_area} = (10 \cdot \text{DS})^2 / 1\text{e}6$ km² per pixel; the whole chain is a pure function of the input scenes and the tunable constants.



Implementation

SAR/

```
|— stac.py          # download: query Planetary Computer STAC, clip to bbox, save GeoTIFFs
|— analyze_water.py # analyse: 3 detectors → figures + georeferenced GeoTIFFs
|— data/Tirupati/   # 16 RTC GRD scenes (VV + VH) (~823 MB)
|— outputs/         # generated PNGs + GeoTIFFs
|— venv/            # Python 3.9.6
```

- **Environment:** venv (Python 3.9.6) — pystac-client, planetary-computer, rasterio, numpy, matplotlib, **scipy** (speckle cleanup), **torch** (SAFNet, optional).
- **Deterministic:** SEED = 0 → identical reruns; a changed result means the code or data changed.
- **Headless:** Matplotlib Agg backend (safe over SSH / CI).
- **Graceful degradation:** guarded imports — without torch the two threshold figures still emit; without scipy speckle cleanup is skipped.
- **Configuration-driven:** all tunables are module-level constants at the top of analyze_water.py; behaviour is changed there, not in the body.
- **Runtime:** ≈ 6 minutes, dominated by SAFNet training on CPU.

Result and Outcome

Surface-water area — the hydrological cycle (dual-pol "confident open water"):

Epoch	Open water
2026-01-06	$\sim 1.4 \text{ km}^2$
spring (drawdown)	declining
2026-06-23 (minimum)	$\sim 0.5 \text{ km}^2$
2026-06-30 (monsoon onset)	$\sim 2.4 \text{ km}^2$ (sharp rebound)

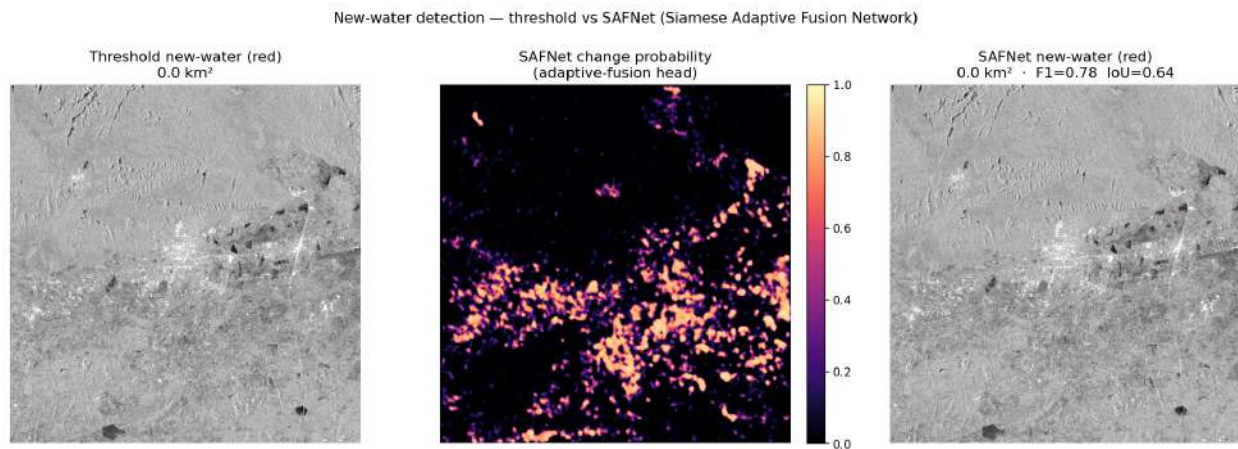
The January→June drawdown as tanks and reservoirs dry, followed by a monsoon-onset refill on the final pass, is itself a **drought-monitoring signal**: a flat or continued decline into June would flag water stress for the district.

New-water (flood) vs dry baseline: $\sim 0 \text{ km}^2$ (16 px by threshold, 25 px by SAFNet). This is **correct** for the window — June water overlaps the *permanent* reservoirs, so almost nothing is *newly* inundated; this is a drawdown period, not a flood event. The dual-pol gate keeps the estimate conservative.

SAFNet vs threshold — scored as a labelled PROXY (not ground truth): precision 0.64 · recall 1.00 · F1 0.78 · IoU 0.64 (SAFNet's 25 px is a superset of the threshold's 16 px).

Outputs (outputs/): water_area_timeseries.png, flood_change_map.png, safnet_change_map.png, plus georeferenced GeoTIFFs new_water_threshold.tif, new_water_safnet.tif, safnet_change_prob.tif (EPSG:32644, QGIS-ready).

Outcome: an operational, reproducible surface-water and flood change-detection capability at ~40 m district scale, with three complementary detectors and GIS-ready products; the 2026 run cleanly captures the district's reservoir drawdown and monsoon-onset refill.



Limitation

- **Dual-pol is deliberately conservative** — reported km² are *confident* open water, not a maximal estimate; thin or wind-roughened water may be undercounted.
- **SAFNet is unsupervised and unvalidated on ground truth** — its precision/ recall/F1/IoU are a **proxy** against the threshold method until a real mask is supplied via REF_WATER; synthetic copy-paste change only approximates real hydrology.
- **The fixed −17 dB cut can be biased** by wind-roughened water and very smooth dry surfaces; the adaptive Otsu curve mitigates this partially, but a DEM/HAND or permanent-water mask would further exclude terrain that cannot hold water.
- **Amplitude-only.** GRD carries no phase, so no interferometric deformation or fine-scale coherence information is available on this track.

References

1. Microsoft Planetary Computer — Sentinel-1 RTC (sentinel-1-rtc) collection.
2. Sentinel-1 mission — ESA / Copernicus Programme.
3. Otsu, N. (1979) — *A threshold selection method from gray-level histograms*, IEEE TSMC.
4. Siamese / attention-fusion change-detection networks for remote sensing (SAFNet design lineage).
5. Bonafilia, D. et al. (2020) — *Sen1Floods11*: a georeferenced dataset for Sentinel-1 flood mapping (ground-truth / future validation).
6. Radiometric Terrain Correction (RTC) γ^0 processing background.